

A Practical Guide to Boosting machine learning

TOPIC -- BOOSTING MACHINE LEARNING

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Status quo for object categorization The recognition of object categories in images is a challenging problem in computer vision, especially when the number of categories is large. This is due to high intra class variability and the need for generalization across variations of objects within the same category. Objects within one category may look quite different. Even the same object may appear unlike under different viewpoint, scale, and illumination. Background clutter and partial occlusion add difficulties to recognition as well. Humans are able to recognize thousands of object types, whereas most of the existing object recognition systems are trained to recognize only a few, e.g. human faces, cars, simple objects, etc. Research has been very active on dealing with more categories and enabling incremental additions of new categories, and although the general problem remains unsolved, several multi-category objects detectors (for up to hundreds or thousands of categories) have been developed. One means is by feature sharing and boosting.

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Problem of object categorization Object categorization is a typical task of computer vision that involves determining whether or not an image contains some specific category of object. The idea is closely related with recognition, identification, and detection. Appearance based object categorization typically contains feature extraction, learning a classifier, and applying the classifier to new examples. There are many ways to represent a category of objects, e.g. from shape analysis, bag of words models, or local descriptors such as SIFT, etc. Examples of supervised classifiers are Naive Bayes classifiers, support vector machines, mixtures of Gaussians, and neural networks. However, research has shown that object categories and their locations in images can be discovered in an unsupervised manner as well.

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Boosting for binary categorization AdaBoost can be used for face detection as an example of binary categorization. The two categories are faces versus background. The general algorithm is as follows:

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There are many boosting algorithms. The original ones, proposed by Robert Schapire (a recursive majority gate formulation), and Yoav Freund (boost by majority), were not adaptive and could not take full advantage of the weak learners. Schapire and Freund then developed AdaBoost, an adaptive boosting algorithm that won the prestigious Gödel Prize. Only algorithms that are provable boosting algorithms in the probably approximately correct learning formulation can accurately be called boosting algorithms. Other algorithms that are similar in spirit to boosting algorithms are sometimes called "leveraging algorithms", although they are also sometimes incorrectly called boosting algorithms. The main variation between many boosting algorithms is their method of weighting training data points and hypotheses. AdaBoost is very popular and the most significant historically as it was the first algorithm that could adapt to the weak learners. It is often the basis of introductory coverage of boosting in university machine learning courses. There are many more recent algorithms such as LPBoost, TotalBoost, BrownBoost, xgboost, MadaBoost, LogitBoost, CatBoost and others. Many boosting algorithms fit into the AnyBoost framework, which shows that boosting performs gradient descent in a function space using a convex cost function.

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Further reading Freund, Yoav; Schapire, Robert E. (1997). "A Decision-Theoretic Generalization of On-line Learning and an Application to Boosting" (PDF). *Journal of Computer and System Sciences*. 55 (1): 119–139. doi:10.1006/jcss.1997.1504. Schapire, Robert E. (1990). "The strength of weak learnability". *Machine Learning*. 5 (2): 197–227. doi:10.1007/BF00116037. S2CID 6207294. Schapire, Robert E.; Singer, Yoram (1999). "Improved Boosting Algorithms Using Confidence-Rated Predictors". *Machine Learning*. 37 (3): 297–336. doi:10.1023/A:1007614523901. S2CID 2329907. Zhou, Zhihua (2008). "On the margin explanation of boosting algorithm" (PDF). In: *Proceedings of the 21st Annual Conference on Learning Theory (COLT'08)*: 479–490. Zhou, Zhihua (2013). "On the doubt about margin explanation of boosting" (PDF). *Artificial Intelligence*. 203: 1–18. arXiv:1009.3613. doi:10.1016/j.artint.2013.07.002. S2CID 2828847.

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Given images containing various known objects in the world, a classifier can be learned from them to automatically classify the objects in future images. Simple classifiers built based on some image feature of the object tend to be weak in categorization performance. Using boosting methods for object categorization is a way to unify the weak classifiers in a special way to boost the overall ability of

categorization.

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Form a large set of simple features Initialize weights for training images For T rounds Normalize the weights For available features from the set, train a classifier using a single feature and evaluate the training error Choose the classifier with the lowest error Update the weights of the training images: increase if classified wrongly by this classifier, decrease if correctly Form the final strong classifier as the linear combination of the T classifiers (coefficient larger if training error is small) After boosting, a classifier constructed from 200 features could yield a 95% detection rate under a 10^{-5} false positive rate. Another application of boosting for binary categorization is a system that detects pedestrians using patterns of motion and appearance. This work is the first to combine both motion information and appearance information as features to detect a walking person. It takes a similar approach to the Viola-Jones object detection framework.

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Implementations scikit-learn, an open source machine learning library for Python Orange, a free data mining software suite, module Orange.ensemble Weka is a machine learning set of tools that offers variate implementations of boosting algorithms like AdaBoost and LogitBoost R package GBM (Generalized Boosted Regression Models) implements extensions to Freund and Schapire's AdaBoost algorithm and Friedman's gradient boosting machine. jboost; AdaBoost, LogitBoost, RobustBoost, Boostexter and alternating decision trees R package adabag: Applies Multiclass AdaBoost.M1, AdaBoost-SAMME and Bagging R package xgboost: An implementation of gradient boosting for linear and tree-based models.

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In machine learning (ML), boosting is an ensemble learning method that combines a set of less accurate models (called "weak learners") to create a single, highly accurate model (a "strong learner"). Unlike other ensemble methods that build models in parallel (such as bagging), boosting algorithms build models sequentially. Each new model in the sequence is trained to correct the errors made by its predecessors. This iterative process allows the overall model to improve its accuracy, particularly by reducing bias. Boosting is a popular and effective technique used in supervised learning for both classification and regression tasks. The theoretical foundation for boosting came from a question posed by Kearns and Valiant (1988, 1989): "Can a set of weak

learners create a single strong learner?" A weak learner is defined as a classifier that performs only slightly better than random guessing, whereas a strong learner is a classifier that is highly correlated with the true classification. Robert Schapire's affirmative answer to this question in a 1990 paper led to the development of practical boosting algorithms. The first such algorithm was developed by Schapire, with Freund and Schapire later developing AdaBoost, which remains a foundational example of boosting.